SAMPLE

Famous Astronomers: Heinrich Olbers

The Paradox of the Dark Sky

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Introduction

Some of astronomy's most consequential figures never held a professional observatory post. The subject of this book earned his living as a physician, spent his days walking the streets of a German port town among patients, and did his observing in the hours in between, from a telescope in the upper story of his own house. Heinrich Wilhelm Olbers, born in 1758 and working long before the word asteroid existed, found two of the first four small planets, rescued a lost one from mathematical oblivion, discovered comets that still carry his name, drew the first serious map of the puzzle that hangs over every dark night, and asked the question that bears his name: why, if the stars go on forever, is the sky dark at night?

The question, now called Olbers' paradox, is the reason his name survives outside the history of astronomy, and it is a good emblem of his whole career. He was a man of evenings, working between obligations, and from that vantage he noticed what the academies had missed: that the darkness overhead is itself an observation, a datum demanding explanation. The paradox stood unresolved for the rest of the nineteenth century, and its eventual answer, in the twentieth, carried information about the universe's history that no one in his era could have imagined. His other discoveries are less philosophical but no less real: the small planets Pallas and Vesta, the recovery of the lost Ceres, a comet-hunting method that professionalized the chase, and a network of younger astronomers he encouraged into careers that reshaped the science.

This book follows the pattern of the series: the discovery of the man and his work, an understanding of what he did and how, his key characteristics, the science behind his observations, the way he observed and measured, his important features and inventions, his context among the astronomers of his age, the modern understanding of his legacy, the research he inspired, his significant discoveries, the challenges and mysteries he faced and left behind, the future of the questions he opened, his trivia, and finally his significance. The account is evergreen: it describes the settled facts of his life and the permanent physics of his questions, so that it ages slowly on the shelf.

A note on the man's double life, since it organizes everything that follows. Olbers belonged to the last generation in which a working physician could also be a first-rank astronomer, and the arrangement was not a hobby but a discipline: he slept little, kept his telescope always ready, taught himself the mathematics of orbits by candlelight, and corresponded with every observatory in Europe. The reader will meet him most often at night, on the roof or at the window of the house in Bremen, between one house call and the next, finding worlds.

A reading note before the history proper. The chapters keep the man's chronology in view but are organized by the series template, so a fact about his boyhood may appear in the chapter on his characteristics and a fact about his last years in the chapter on his science; the index of the book is the chapter list itself, and the reader wanting pure chronology may read the first two chapters and the last two in sequence. The documentation behind the account is the standard record of the era: his published papers and comet method, the great edition of his correspondence, the memoirs written by the astronomers who knew him, and the modern histories that have sifted all of it. Nothing speculative is presented as fact, and the era's uncertainties are marked as such in the text, which is the only way a two-century-old life can be honestly told.

One more paragraph of the history belongs to its close: the end of the beginning. By the first decade of the nineteenth century the physician of Bremen had become, in the small world of European astronomy, an institution: the man whose comet orbits were trusted, whose letter could settle a priority dispute, whose roof was on the tour of every traveling observer, and whose recommendation could carry a young man into the profession. Nothing about the institution was official; no court had appointed him and no academy owned him. The history's next chapters, of the finds and the theory and the paradox, all rest on this one, and this one rests on twenty years of evenings, spent exactly as described, before anyone outside the network had heard his name; the reader should mark the delay between the work and the fame, since the delay is the rule of the scientific life and never more visible than in his.

A final paragraph of the history sets the dates in one line, as the record keeps them. Born 1758 in the village near Bremen; schooled in the town; trained in medicine at Gottingen, with the eye's anatomy for his dissertation; returned to Bremen in 1781 and began the practice; observing seriously from the 1770s, publishing from the 1780s; the great years of the small planets from 1802 to 1807; the comet of 1815; the paradox paper in 1823; the letters, the mentorships, and the observatory through the middle decades; observing into his last year, and dead in Bremen in 1840, in his eighty-second year, a few months after his last recorded observations. The line is the life; the chapters that follow are its reasons.

Chapter 1: The Discovery of Heinrich Olbers

The life begins in the countryside near the free city of Bremen, in a parsonage, in the middle of the eighteenth century.

Heinrich Wilhelm Matthias Olbers was born in the village of Arbergen, just east of Bremen, in 1758, the son of a pastor. The boy's schooling in Bremen showed the pattern early: he learned the classics and the languages well enough, but it was the sky that claimed him, and by his own account he observed a comet at seventeen with a determination that decided his path. There was no career in astronomy open to a pastor's son of no independent means, and the family's practical choice was medicine, the learned profession that could support a man and leave his evenings his own.

He studied medicine at Gottingen, and the university years fixed his double life deliberately. His teacher there was the physicist who first measured the shape of the Earth's magnetic field, and the medical training included the physics of light and the mathematics of motion; Olbers took both, and wrote his dissertation on a question of the human eye, a medical study of the anatomy of the orbit that foreshadowed nothing less than a lifetime of looking. He kept observing through the whole of his study, made his first comet observations, and left the university a physician who understood optics and orbits as well as most astronomers of the day.

He went back to Bremen in 1781 and began the medical practice that would occupy him for the next six decades. The town took him quickly: he was skilled, unhurried, and famously devoted to the poor, among whom he rode at all hours, and his reputation as a healer grew to the point where his evenings, the astronomer's working hours, were protected by the town's own affection for its doctor. He never left Bremen again, and Bremen never let him go; the whole astronomical career that follows was conducted from one town, one house, and one telescope.

The discovery of Olbers as an astronomer, in the public sense, came in the last years of the eighteenth century. His comet observations and his skill at computing orbits had made him known in the small European network of working astronomers, and when the century turned and the great puzzle of the missing planet appeared, the network turned to him. The story belongs to a later chapter; the fact belongs here: by 1800 the physician of Bremen was one of the two or three people in Europe whom the professionals trusted with the mathematics of the sky.

A word about his character, since every later chapter depends on it. The documentary record is unusually consistent: his correspondents describe a man of boundless generosity and no patience for priority disputes, who gave his methods away freely, spent his nights checking other people's observations, and treated the young astronomers who visited him as colleagues from the first letter. He was, by the testimony of everyone who wrote about him, the most helpful man in the science, and his house in Bremen became a station on every astronomical tour of Europe, where the callers, from the young mathematician Gauss to the traveler visiting on business, found the door open, the telescope ready, and the doctor delighted.

The discovery of Olbers, then, was the discovery the town made and the profession confirmed: that the evening hours of one physician, systematically spent, could do the work of an observatory, and that the dark sky over Bremen, watched by the right person, was as productive as any hilltop in Europe.

A note on the instruments of his understanding completes the chapter. The telescope of his youth was a small refractor, a spyglass by modern standards; his middle years saw the acquisition of better glasses, including a fine refractor he worked for decades; and his observatory years had the house above the stable with its mounting and clock. But the reader should understand that the glass was never the point: the era's discoveries were made by position measurement and arithmetic, and the tools that found Pallas and Vesta were a micrometer, a clock, a table of logarithms, and the patience to use all three at two in the morning. The telescope gathered the light; the observer, and this observer above all, did the rest.

A last note on the understanding of the man: his two careers shared a habit of mind that the reader can see in both. The doctor's method is the astronomer's method: observe carefully, record exactly, compare the record with what is known, and revise when the case requires; his medical contemporaries noted his refusal to theorize ahead of symptoms, and his astronomical correspondents noted the same refusal about orbits and observations. The habit is why his wrong theories, judged by the standards of his era, were respectable: they were the best readings of the symptoms available, offered with the case left open for review. A physician to the last in both professions, he never closed a case he had not examined, and he kept the whole sky as an open file.

A note on the medicine's astronomy completes the chapter's account. The era's physicians were trained in

mathematics and natural philosophy as part of the degree, and the training was not ornamental: the oculist's refractions, the apothecary's weights, and the astronomer's reductions all ran on the same arithmetic and the same instruments, the rule, the table, and the clock. His two professions were therefore closer than their modern descendants would guess, twin crafts of careful measurement under responsibility, and the reader who wonders how a doctor found the discipline for the roof should credit the discipline's first home, the sickbed, which taught the same lessons earlier: show up, pay attention, write it down, and do not guess.